Business Model Design

Project Name: VitalBee
Team Name: DİH

Task 1: Answer key business model questions

1. How do you acquire customers?
We plan to acquire customers by working with government-assisted disaster response organizations and public institutions. We will participate in government-supported R&D programs, build partnerships with public agencies, and use these relationships to introduce our solution to potential institutional buyers.

2. After you have landed a new customer, how do you plan to relate to that customer and manage the relationship (if at all)?
We plan to maintain long-term relationships through onboarding and training programs, technical support, regular maintenance, and continuous software updates. Since our product is mission-critical, customer trust and long-term service quality are very important.

3. How do you charge your customers? What is your revenue model?
We use a hybrid revenue model consisting of a one-time hardware sale, annual software subscriptions, and optional maintenance and training services. Customers can purchase our drone kits through procurement contracts and continue using our software platform through yearly subscriptions.

4. How much do you charge your customers? Can you calculate your revenues for the next month, quarter and year?
Our pricing assumptions are as follows:

- Drone kit price: **$15,000 per unit**
- Annual software subscription: **$3,000 per unit**
- Optional maintenance and training: **$2,000 per client per year**

Based on our early-stage sales assumptions, our estimated revenues are:

- **Next month: $16,250**
- **Next quarter: $48,750**

- **Next year: $205,000**

These estimates are based on limited early customer adoption and are expected to grow as more institutions adopt the system.

5. What assets are available to you or under your control?
Our key assets include our intellectual property, academic know-how, technical expertise in deep learning and anomaly detection systems, and our research network. Our prior work in this field and our access to academic and technical guidance are strong assets that support product development.

6. Who are your key partners?
Our key partners are STM Defence and Aselsan for sensor and hardware support, as well as NVIDIA and Anthropic for AI and computing support. These partners can help us improve sensing, signal processing, and reliable AI-based decision-making.

7. What key activities do you need to engage in to deliver your value proposition?
Our key activities include payload optimization, autonomous flight planning, signal processing, AI model development, probability map generation, dashboard development, pilot testing, and regulatory preparation. These activities are necessary to build and deliver a reliable survivor detection system.

8. What are your fixed costs?
Our fixed costs mainly include personnel salaries, software development expenses, subscriptions for engineering tools, and essential infrastructure costs. These are the regular costs required to continue operating and developing the system.

9. What are your variable costs? Can you calculate your total cost for the next month, quarter and year?
Our variable costs include drone production or procurement, payload integration, onboard computing components, field testing expenses, certification processes, deployment-related maintenance, and scaling-related operational costs.

Based on our current planning assumptions, our estimated total costs are:

- **Next month: $25,000–$50,000**
- **Next quarter: $40,000–$90,000**

- **Next year: $80,000–$180,000**

These amounts may vary depending on prototype development, certification progress, testing intensity, and production scale.

10. Does your revenue forecast demonstrate increased profitability towards the end of the forecast period?
Yes. Our revenue model combines one-time hardware sales with recurring annual software subscriptions and optional service revenue. This structure creates stronger long-term revenue potential, so profitability is expected to improve over time as the customer base grows and recurring income increases.

Task 2: Create your own business model

The Business Model Canvas

Designed for: Disaster Response Operations
Designed by: DİH
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Key Partnerships

- STM Defence and Aselsan for sensor, hardware, and defense-grade technical support
- NVIDIA for AI computing infrastructure and edge processing support
- Anthropic or similar AI partners for advanced AI model support
- Universities, academic advisors, and research labs for technical validation and R&D guidance
- Government-supported disaster response institutions and public agencies for pilot programs and field collaboration
- Certification and regulatory bodies for compliance and deployment approval

Key Activities

- Drone and payload design optimization
- Autonomous flight planning and area scanning
- Signal collection and processing
- AI model development for survivor detection
- Multi-sensor data fusion and probability map generation
- Dashboard and software platform development
- Pilot testing in realistic disaster environments
- Certification, regulatory preparation, and deployment planning
- Maintenance, updates, and customer training

Key Resources

- Intellectual property and technical know-how
- AI models and signal-processing expertise
- Drone system design and prototype capability
- Academic knowledge and research support
- Software platform and analytics dashboard
- Sensor integration capability
- Engineering team and development expertise
- Institutional network and public sector connections

Value Propositions

- Faster survivor detection in disaster areas
- Reduced search time for rescue teams
- Safer and more efficient operations in risky environments
- AI-supported decision-making instead of only manual search
- Accurate scanning in complex, low-visibility, and debris-filled areas
- A scalable and modular platform that can improve with new sensors and software updates
- Better resource allocation for emergency response teams

Customer Relationships

- Drone and payload design optimization
- Autonomous flight planning and area scanning
- Signal collection and processing
- AI model development for survivor detection
- Multi-sensor data fusion and probability map generation
- Dashboard and software platform development
- Pilot testing in realistic disaster environments
- Certification, regulatory preparation, and deployment planning
- Maintenance, updates, and customer training

Channels

- Government-supported R&D programs
- Public procurement processes
- Direct partnerships with disaster response institutions
- Pilot projects and field demonstrations
- Academic and technical collaborations
- Industry events, disaster technology exhibitions, and public-sector meetings
- Direct sales to institutions and agencies

Customer Segments

- Government-assisted disaster response organizations
- Public emergency management institutions
- Search and rescue teams
- Civil defense organizations
- Municipalities and public safety agencies
- Large humanitarian and disaster-relief organizations

Cost Structure

Fixed Costs

- Personnel salaries
- Software development expenses
- Engineering tools and software subscriptions
- Core infrastructure and office/lab costs
- Administrative and operational expenses

Variable Costs

- Drone production or procurement
- Payload and sensor integration
- Onboard computing hardware
- Field testing and simulation costs
- Certification and regulatory costs
- Maintenance and deployment-related costs
- Scaling and production expenses

Revenue Streams

- One-time drone kit sales
- Annual software subscription fees
- Optional maintenance and training services
- Future premium analytics or advanced reporting services
- Possible public-sector contracts and long-term service agreements

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Task 3: Consider your business model

1. Does it work? Do the revenues outweigh the cost? Try estimating revenues and cost for the next month (or the first month of sales, if you are without revenue for now)

We assume the business will not be profitable for the early stages. Our main cost will be the R&D expenses throughout the development stages. We aim to sell one drone with the annual subscription package. This will generate $18,000 ($15,000 + $3,000) for the first month. We are anticipating $12,500 variable cost ($6,000 hardware + $4,000 sensor + $2,500 assembly and testing) for 1 drone alone in the first month. The salaries and software infrastructure will cost around $10,000.
Total cost in first month: $12,500 + $10,000 = $22,500
Total revenue in first month: $15,000 + $3,000 = $18,000
Even with an optimistic assumption of 1 unit of hardware sale and 1 unit of annual subscription sale in the first month, we would lose $4,500. The first month is not profitable but this is expected because hardware costs are high and subscription base is still small. As the company gets bigger and we will be able to make bulk purchases of sensors and hardware thus we will produce much more product with lower cost rates and we will sell more subscriptions and increase our subscription base and business model will become profitable in less than a year.

2. What are the risks to your business model? What parts of your business model are most critical for your business to grow in a profitable manner?

We expect strong competition in the search and rescue field. This is a very serious risk for a startup, as established companies already have proven technologies, existing customer networks, and easier access to procurement channels. Competing against such players may make market entry and customer acquisition more challenging, especially in the early stages. Additionally, we are planning to do business with government institutes, but government procurement processes are typically slow and bureaucratic. This situation can result in long delays in revenue generation and may slow down the company's growth momentum. Delayed contracts and payments can also create cash flow challenges, which are critical for a startup operating with limited resources. Another important risk is the need for extensive field validation. Since our product operates in mission-critical disaster scenarios,

customers will require strong evidence of reliability and accuracy before adoption. Without successful pilot deployments and proven performance, it may be difficult to build trust and secure contracts.

3. Are there things that can or should be changed to strengthen the business model or reduce its risk?
To strengthen the business model and reduce risk, we can shift part of our focus from one-time hardware sales toward a more recurring, service-based model. Introducing options such as leasing or a “drone-as-a-service” model would lower the upfront cost for customers and make adoption easier, especially for public institutions with limited budgets. At the same time, expanding our software subscription tiers with advanced analytics, reporting, and AI features would increase recurring revenue and improve profit margins. This transition would reduce dependence on hardware sales and create a more stable and predictable income stream over time.

Task 4: Clarify your business model assumptions (cont'd)

No.	Questions	Fact (with evidence)	Assumption
1.	Who are our Key Partners? Who are our key suppliers? Which Key Resources are we acquiring from partners? Which Key Activities do partners perform?	• We identified STM Defence and Aselsan as key partners for sensor and hardware support, since they provide *specialized sensor technologies, antenna design know-how, and ruggedized hardware components.* • We rely on AI and computing partners such as NVIDIA because *edge computing platforms and AI acceleration hardware are essential for AI model training and inference.* • We collaborate with universities and research labs for validation, as our project includes *controlled pilot tests and academic research support.* • Public institutions act as partners through pilot deployments and co-funded trials.	• We assume these partners will be willing to collaborate with a student-led startup. • We assume access to defense-grade hardware will be feasible and affordable. • We assume public institutions will participate in pilot programs without major bureaucratic barriers.
2.	What Key Activities do our Value Propositions require?	• Our system requires AI model development, signal processing, and sensor fusion, since the platform *combines AI-based detection algorithms and multi-sensor fusion.* • We must perform autonomous flight planning and scanning, as the drone *performs wide-area scanning through an adaptive grid-based search algorithm.* • Field testing and validation are essential because the project includes *controlled pilot tests with realistic debris setups.* • We need dashboard and software platform	• We assume we can successfully integrate all components (AI + hardware + drone) without major technical bottlenecks. • We assume field testing environments will be accessible. • We assume our small team can handle all critical development activities initially.

		development to deliver results in real time.	
3.	What Key Resources do our Value Propositions require?	• Our core resources include AI models, signal processing, and software systems, since all fusion and scoring software is *"developed in-house".* • We depend on drone hardware and sensors, as the system integrates UAVs, RF, acoustic, and thermal sensing. • Our team expertise is a key resource, as each member contributes to AI, system architecture, and signal processing.	• We assume our technical knowledge is sufficient to build a working prototype. • We assume we can maintain and improve these systems without large external teams. • We assume funding (grants, etc.) will be secured to access required hardware.
4.	What value do we deliver? Which problems do we solve?	• We solve the problem of slow and inefficient rescue by enabling *rapid survivor localization in post-disaster environments.* • Our system improves decision-making by providing *a probability heatmap with ranked priority points instead of raw sensor data.* • We reduce risk because traditional methods are *too slow, dangerous, or limited in coverage.*	• We assume rescue teams will trust AI-assisted recommendations. • We assume our system will significantly outperform existing solutions. • We assume customers value decision-support more than raw data.
5.	What type of relationship do customers expect?	• Our customers are institutional (government, SAR teams), which implies structured and long-term relationships. • Our model includes training, maintenance, and support services.	• We assume customers expect high-touch, long-term support. • We assume trust and reliability are more important than price. • We assume ongoing support will be necessary due to mission-critical usage.
6.	Through which channels do customers want to be reached?	• We reach customers via pilot projects and public-sector collaborations. • The market already uses institutional procurement channels, as public agencies actively use drones.	• We assume direct sales and partnerships are more effective than online channels. • We assume demonstrations and field tests are the most persuasive channel.

			• We assume procurement processes will remain the primary sales channel.
7.	What are the most important costs?	• Major costs include drone hardware, sensors, and AI development, as prototype funding is allocated to these areas. • Testing, certification, and deployment are also key cost drivers.	• We assume hardware costs will decrease with scale. • We assume software costs will stabilize over time. • We assume certification costs will not exceed our budget significantly.

8.	For what value are customers willing to pay?	• Our revenue comes from:
		  ○ Drone sales
		  ○ Annual software subscriptions
		  ○ Maintenance and training services
		• Customers currently pay for separate drone systems and detection tools, which are less integrated.
			• We assume customers are willing to pay a premium for integrated solutions.
			• We assume subscription-based software will be accepted in public sector contracts.
			• We assume recurring revenue will grow over time and dominate total revenue.

Task 5: Create a Value Proposition Canvas

The Value Proposition Canvas

Value Proposition:

Customer Segment:

Products and Services

- AI-powered autonomous drone system
- Multi-sensor payload (RF, acoustic, thermal, etc.)
- Survivor detection and probability mapping software
- Autonomous flight planning and area scanning
- Post-mission analytics and reporting
- Training, maintenance, and support services

Gain Creators

- Enables much faster detection of survivors
- Provides actionable insights instead of raw data
- Improves rescue success rates
- Allows better resource allocation and planning
- Enhances operational efficiency in large-scale disasters

Pain Relievers

- Automating search and reduces manual effort and risk
- Using multi-sensor fusion works in low-visibility conditions
- AI-based probability maps eliminates guesswork
- Autonomous scanning ensures systematic area coverage
- Reducing false positives by combining multiple signals
- Minimizes time wasted on low-priority areas

Gains

- Faster and more accurate survivor detection
- Clear prioritization of high-probability rescue locations
- Reduced operational risk for rescue teams
- Better decision-making with reliable data
- Efficient use of time and limited resources

Customer Jobs

- Scan large and complex areas efficiently
- Allocate rescue teams and resources effectively
- Operate safely in dangerous environments (unstable structures, debris

Pains

- Manual search is slow, risky, and labor-intensive
- Limited visibility in debris, dust, and low-light environments
- Existing tools (thermal cameras, dogs, etc.) are not always reliable alone
- Risk of missing survivors due to incomplete coverage

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